Temporary title

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1. **Order and Degree**
   In general we have to square till the equation is a polynomial in the derivatives of the dependent variable (which does not include the zeroth derivative)

   (a) Order 2 Degree 2
   (b) Order 2 Degree 2
   (c) Order 2 Degree 2
   (d) Order 2 Degree 2
   (e) Order 2 Degree 3

2. **ODE formation**
   Here, we will first try to classify this as one of the standard classes of differential equations, failing that we differentiate the equation till each of the arbitrary constants is eliminated

   (a) **Critical damping equation**

   $$y'' - 2ky' + k^2y = 0$$

   (b) **Differentiate, invert and differentiate**

   $$\begin{aligned} y &= \ln(\sin(x+a)) + b \\ y' &= \cot(x+a) \\ \cot^{-1} y' &= x + a \\ -\frac{1}{1+(y')^2} y'' &= 1 \\ y'' + (y')^2 + 1 &= 0 \end{aligned}$$

   $$werecognisethisasafirstorderdifferentialequationin$$

y'$oftheBernoullitype$

(c) **A first order ODE**

$$y = xy' + y' - (y')^3$$

(d) **Shift a known equation**
First we eliminate a to get

$$y' = -\frac{x-h}{y-k}$$

Now we eliminate h

$$y''(y-k) + (y')^2 = -1$$

Now we eliminate k by first bringing it out

$$y - k = -\frac{1+(y')^2}{y''}$$

and differentiating

$$-y' = \frac{2y'(y'')^2 - y'''(1+(y')^2)}{y''^2}$$

or

$$3y'(y'')^2 = y'''(1+(y')^2)$$

which is again a second order differential equation in y'

(e) **More circles**